MultiCore TECHNOLOGIES

## LEARN TODAY BUILD TOMORROW

Your Future in Tech

# DATA SCIENCE WITH AI

## MASTER PROGRAM

Turn Data into Insights,
Build Intelligent Solutions




## COURSE HIGHLIGHTS

- Python Programming & Data Science
- Statistics & Exploratory Data Analysis
- Machine Learning & Model Development
- Artificial Intelligence & Deep Learning
- SQL, Power BI & Real-World Projects

# DATA SCIENCE WITH GEN AI

Learn • Build • Analyze • Create

## COURSE OVERVIEW

A practical, career-focused program covering Python programming, data analysis, statistics, machine learning, artificial intelligence, deep learning, SQL, and Power BI to develop end-to-end data science skills.

## WHAT WILL YOU LEARN?

### PYTHON PROGRAMMING

- Python fundamentals
- Advanced programming concepts
- Functions, modules & libraries
- Exception & file handling
- NumPy, Pandas, Matplotlib, Seaborn & SciPy

### DATA SCIENCE

- Data cleaning & preprocessing
- Exploratory Data Analysis
- Data transformation
- Visualization & statistical analysis
- Data aggregation, merging & joining

### MACHINE LEARNING

- Supervised & unsupervised learning
- Regression & classification
- Clustering & dimensionality reduction
- Model evaluation
- Feature selection & hyperparameter tuning

### AI & DEEP LEARNING

- AI concepts & search algorithms
- Natural Language Processing
- Computer Vision
- Neural Networks
- CNN, RNN, GAN & Autoencoders

### DATA ANALYTICS

- SQL querying & database concepts
- Advanced SQL
- Power BI dashboards
- Data modeling & Power Query




## WHO CAN JOIN?

- Students & graduates
- Beginners in programming
- Aspiring Data Scientists
- Aspiring AI/ML Engineers
- Data Analysts
- Career changers
- Working professionals
- Analytics & technology enthusiasts

## LEARNING OUTCOMES

**By the end of the course, learners can:**

- Write Python programs for data science
- Clean, transform & analyze datasets
- Perform EDA and create visualizations
- Apply machine learning algorithms
- Evaluate and optimize ML models
- Work with AI, NLP & Computer Vision
- Build deep learning models
- Query and analyze data using SQL
- Create interactive Power BI dashboards
- Develop end-to-end data science projects

SKILLS TODAY
FOR A SMARTER TOMORROW

# PYTHON, DATA SCIENCE & STATISTICS

**Build real-world skills in Python, data science and statistics for smarter decisions and better solutions**



- PRACTICAL LEARNING
- REAL-WORLD PROJECTS
- INDUSTRY-RELEVANT SKILLS
- BUILD A STRONG DATA SCIENCE FOUNDATION

**TURN DATA INTO INSIGHTS**

## PYTHON PROGRAMMING

### PYTHON FUNDAMENTALS

- Variables, data types & operators
- Conditional statements
- For & while loops
- Functions, modules & libraries
- Exception handling
- File handling

### PYTHON DATA SCIENCE LIBRARIES

**NumPy**

- Arrays
- Matrix operations
- Vectorized computation

**Pandas**

- Series & DataFrames
- Data manipulation
- Data analysis

**Matplotlib & Seaborn**

- Data visualization
- Statistical plots
- Heatmaps
- Pair plots

**SciPy**

- Scientific computing
- Optimization
- Linear algebra

## DATA SCIENCE WITH PYTHON

### DATA CLEANING & PREPROCESSING

- Missing data
- Imputation techniques
- Data transformation
- Normalization & scaling
- Categorical encoding
- Outlier handling

### EXPLORATORY DATA ANALYSIS

**EDA**

- Descriptive statistics
- Distribution analysis
- Correlation analysis
- Data aggregation & grouping

### DATA VISUALIZATION

- Histograms
- Distribution plots
- Box plots
- Statistical visualizations

### DATA MANIPULATION

- DataFrame merging
- One-to-one relationships
- One-to-many relationships
- Many-to-many relationships
- Concatenation & appending

## STATISTICS FOR DATA SCIENCE

### STATISTICS FUNDAMENTALS

- Mean, median & mode
- Variance & standard deviation
- Probability basics
- Probability distributions

### STATISTICAL ANALYSIS

- Normal distribution
- Binomial distribution
- Sampling
- Central Limit Theorem

### HYPOTHESIS TESTING

- p-value
- T-Test
- Chi-Square Test
- Correlation & covariance

### STATISTICS WITH PYTHON

- NumPy
- Pandas
- SciPy
- Statistical analysis for EDA & ML

## KEY SKILLS

Collect → Clean → Explore → Visualize → Analyze

# MACHINE LEARNING & ARTIFICIAL INTELLIGENCE

## MACHINE LEARNING

### SUPERVISED LEARNING

**REGRESSION**
- Linear Regression
- Polynomial Regression

**CLASSIFICATION**
- Logistic Regression
- K-Nearest Neighbors
- Decision Trees
- Support Vector Machines
- Random Forest

**MODEL EVALUATION**
- Accuracy
- Precision
- Recall
- F1-Score
- ROC & AUC

### UNSUPERVISED LEARNING

**CLUSTERING**
- K-Means
- Hierarchical Clustering

**DIMENSIONALITY REDUCTION**
- PCA
- t-SNE

**ASSOCIATION**
- Apriori
- Eclat

### MODEL SELECTION
- Cross Validation
- Grid Search
- Hyperparameter Tuning
- Overfitting & Underfitting

## ARTIFICIAL INTELLIGENCE

### AI FUNDAMENTALS
- AI concepts
- Problem solving
- Intelligent search

### SEARCH ALGORITHMS
- A*
- Minimax
- Alpha-Beta Pruning

### NATURAL LANGUAGE PROCESSING

**TEXT PROCESSING**
- Tokenization
- Lemmatization
- Stopword removal

**NLP APPLICATIONS**
- Text classification
- Sentiment analysis
- Named Entity Recognition

### COMPUTER VISION
- Image classification
- Object detection
- Convolutional Neural Networks

### KEY SKILLS

Prepare → Train → Evaluate → Optimize → Predict

BUILD INTELLIGENT SOLUTIONS

EXPLORE REAL-WORLD PROBLEMS

GAIN HANDS-ON EXPERIENCE

BOOST CAREER GROWTH

**SKILLS TODAY**
FOR A SMARTER TOMORROW

# DEEP LEARNING, SQL, POWER BI & PROJECT

**BUILD SKILLS + WORK ON REAL DATA + CREATE IMPACT**

## DEEP LEARNING

### NEURAL NETWORK FUNDAMENTALS
- Perceptron
- Activation functions
- Backpropagation
- Gradient Descent

### DEEP LEARNING MODELS
- FNN – Feedforward Neural Networks
- CNN – Image processing, Computer vision tasks
- RNN – Time-series forecasting, Text generation
- GAN – Generating new data
- Autoencoders – Anomaly detection

### DEEP LEARNING FRAMEWORKS
- TensorFlow / Keras
- Model building & training
- PyTorch
- Dynamic computational graphs
- Model deployment

### TRANSFER LEARNING
- Pre-trained models
- Fine-tuning
- ImageNet models

## SQL & DATABASE

### SQL FUNDAMENTALS
- SELECT
- WHERE
- ORDER BY
- LIMIT
- DISTINCT
- COUNT, SUM, AVG
- MIN & MAX

### ADVANCED SQL
- INNER, LEFT, RIGHT & FULL JOIN
- GROUP BY & HAVING
- Subqueries
- Window Functions
- ROW_NUMBER & RANK
- CASE Statements

### DATABASE CONCEPTS
- Normalization
- Denormalization
- Indexing
- Views
- Triggers
- Data aggregation



## POWER BI

### DATA ANALYTICS & VISUALIZATION
- Power BI Desktop
- Power Query
- Data transformation
- Data modeling
- Reports & dashboards

### POWER BI SERVICE
- Publishing & sharing
- Collaboration

## PROJECT & FINAL INTEGRATION

### BUILD A COMPLETE DATA SCIENCE & AI PROJECT

**PROJECT JOURNEY**

Collect → Clean → Analyze → Model → Visualize → Evaluate → Deploy

### PROJECT COMPONENTS

#### DATA
- Data cleaning
- EDA
- Statistical analysis

#### MACHINE LEARNING
- Model development
- Model evaluation
- Hyperparameter tuning

#### AI / DEEP LEARNING
- NLP / Computer Vision
- Neural network models
- AI-powered applications

#### BUSINESS INTELLIGENCE
- SQL analysis
- Power BI dashboards
- Data-driven insights

## FINAL OUTCOME

Code • Analyze • Predict • Visualize • Build

## TECHNOLOGY STACK

Python | NumPy | Pandas | Matplotlib | Seaborn | SciPy | Scikit-learn | TensorFlow | Keras | PyTorch | SQL | Power BI

**>>> BECOME DATA-READY. AI-READY. CAREER-READY. <<<**

- PRACTICAL LEARNING
- REAL-TIME SCENARIOS
- HANDS-ON PROJECTS
- CAREER GROWTH

**SKILLS TODAY FOR A SMARTER TOMORROW**